

Tyler Baker

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Portfolio: <https://mtbty43.github.io>

Mechanical engineer with hands-on experience in UAVs, R&D, and advanced manufacturing, managing projects through design, fabrication, and field deployment under schedule and customer constraints.

Education

B.S. Mechanical Engineering, Engineering Management Minor, FE/EIT Cert.

May 2024

Western Colorado University (WCU) - CU Boulder Engineering Partnership Program

GPA: 3.66

Work Experience

R&D Engineer

Aug. 2024 - Feb. 2026

Western Colorado University/Rocky Mountain Biological Laboratory (Gunnison, CO)

- Designed and deployed ground/UAV field-based sensor and mechanical systems for high-alpine environments across 3 deployments, accounting for power, cost, and field deployment constraints
- Evaluated sensor system performance utilizing field data to inform iterative design revisions and in one case decreasing station failure rate by >75%
- Contributed custom sensor and data collection systems in support of 3 scientific publications, 1 M.S. thesis, 1 PhD dissertation, and 2 hydrology projects

Drone Engineer Independent Contractor (Part Time)

Jan. 2025 – Feb. 2026

Ghost Army Engineering (Gunnison, CO)

- Performed UAV assembly, integration, maintenance, and field operations for an early-stage aerospace startup, supporting 10-20 individual flight tests contracting under my own LLC
- Independently diagnosed and resolved in-field technical issues under evolving requirements, increasing flight readiness and capabilities across multiple drone fleets

Mechanical Engineering Internship

May - Aug. 2023 & 2024

Blister Labs/Western Colorado University (Gunnison, CO)

- Utilized custom load and impact testing fixtures to evaluate compliance and damping characteristics of 7 mountain bike wheel models under controlled loading scenarios
- Integrated sensor and motion capture data to correlate experimental testing with real-world riding data and compare wheel designs and materials to inform rider wheel selection

Lead Makerspace Teaching Assistant

Jan. 2021 – May 2024

Western Colorado University (Gunnison, CO)

- Led daily makerspace operations including training, supervising, and process improvement projects

Projects

SAE Baja Vehicle and Competition (Senior Capstone)

Aug. 2023 – May 2024

- Led as a systems engineer to design, fabricate, and evaluate a custom off-road vehicle
- Contributed CAD modeling, manufacturing planning, fabrication, documentation, and project management under strict cost, time, and competition constraints

Adaptive Gravel Bike (Component Design Project)

Aug. 2022 – May 2023

- Collaborated with the Crested Butte Adaptive Sports Center and industry partners to design and build an ultralight bike frame utilizing SLS titanium additive manufacturing and bonded carbon fiber
- Applied failure mechanics principles and rider-specific load cases to inform structural design requirements while balancing strength-to-weight and user needs

Custom Steel Gravel Bike Frame (Independent Study)

Aug. 2023 – Dec. 2023

- Fabricated a custom fit gravel bike frame utilizing machining, TIG welding/sequencing, tube bending/notching, and fixturing techniques

Technical Skills & Certifications

- Manufacturing: TIG/MIG Welding, Sheet Metal Fabrication, CNC & Manual Machining, Additive Manufacturing (FDM, SLA, SLS), Composite Bonding, Fixturing
- Software & Tools: SolidWorks (CSWA), Fusion 360, MATLAB, Python, Arduino IDE
- Process: Project Management (CPO), Lean/Six Sigma Principles, DFM/DFA, GD&T, FEA

Clubs/Organizations

- SAE Baja - Team Captain, ASME, WCU Mountain Bike Team